onboard computing. As a result, flight stability depended entirely on the operator’s skill, weather conditions, and visual tracking capabilities.

While these aircraft lacked any form of automation or sensory feedback, their experiments laid the groundwork for key UAV technologies that would emerge later such as autopilot systems, telemetry, and sensor integration. The Aerial Target (1916) and Queen Bee (1935) exemplified the application of these foundational technologies, combining basic radio control with powered propulsion and fixed-wing designs.

The technological foundations of first-generation drones reflect a fusion of mechanical ingenuity and emerging electronic control, driven by early experimentation in wireless communication and aeronautical engineering. These pioneering systems established the core technological architecture propulsion, control, communication, and structure that remains central to all UAV designs today. For students and researchers, studying these foundational technologies provides valuable insight into how simple mechanical systems evolved into the complex, intelligent, and autonomous drones that define modern aerial engineering.

2.4.1 Principles and components of first-generation drones

The first generation of drones was built upon fundamental principles of remote control, aerodynamics, and mechanical propulsion. Although technologically primitive, these early UAVs demonstrated that controlled, pilotless flight was achievable through coordinated systems of communication, propulsion, and structural design. Their operation depended entirely on human-controlled inputs, transmitted via basic radio signals, and executed through mechanical linkages connected to flight surfaces.